Mini Projects - SMRTTECH 4SC3

Smart Cities Simulation, Analysis, and Digital-Twin Showcase



Project Overview

Students will work in groups of three or four and act as a smart transportation consulting team. Each group will create a fictional consulting-company name and will complete two separate client projects during the term.

Mini Project A - Roadway Traffic Simulation and Analysis (SUMO)

Mini Project B - Railway Digital Twin and Immersive Visualization (Blender and Unity)

The two mini projects are independent engagements. Mini Project B does not require students to recreate the roadway study area from Mini Project A. Together, the two projects demonstrate how simulation, engineering analysis, and digital twins can support smart-city transportation planning across both roadway and railway modes.

Each mini project has its own staged submissions, its own technical report, and its own project website. A report template will be provided for each mini project.

Deliverables and Deadlines

Deliverable	Content	Due	Weight
Mini Project A - Stage 1	SUMO road network of the assigned study area	Oct 5	**5%**
Mini Project A -	Performance analysis + Report A	Oct 19	**10%**

Deliverable	Content	Due	Weight
Stage 2			
Mini Project B - Stage 1	Railway digital-twin assets (Blender)	Nov 2	**5%**
Mini Project B - Stage 2	Unity VR environment + Report B	Nov 20	**10%**
Final Presentation	Two project websites, LinkedIn post, presentation	Dec 4	**10%**

Late submissions are subject to the course late policy. All group members are expected to contribute to the technical work, engineering analysis, reports, websites, and final presentation.

Mini Project A - Roadway Traffic Simulation and Analysis

Client brief: Autonomous vehicles are expected to play an increasingly important role in urban transportation. In this project, your consulting team will first establish a SUMO roadway-and-traffic baseline, then compare a 100% human-driven vehicle (HDV) condition with a 100% autonomous vehicle (CAV) condition and evaluate one selected intervention.

Your consulting team has been hired for a hypothetical City of Toronto case study. The team will develop the assigned roadway network in Stage A1, configure traffic and run three controlled conditions in Stage A2, and communicate thve integrated investigation in Report A.

Objectives of Mini Project A:

- Develop a digital roadway network and traffic model for the assigned Toronto study area and characterize the 100% HDV baseline.
- Compare the selected edge's Speed–Density relationship for the 100% HDV baseline and 100% CAV scenario.
- Select, justify, implement, and evaluate one intervention using Speed–Density evidence on the same selected edge.

Stage A1 - Urban Roadway Simulation in SUMO

Assigned Study Area

Each group has been assigned a study area in the City of Toronto. Use the GIS map provided for your group as the reference for developing your SUMO road network.

The study area includes signalized intersections, major roads, and secondary and local roads, providing sufficient roadway complexity for the project.

SUMO Network Development

Each group must develop a SUMO model that reasonably represents the selected study area. You should reasonably apply Milestones 1, 2, and 3 in Lab 1. The network should include:

- Roadway and junction geometry that reasonably matches the provided GIS map.
- Roadway lanes and turning lanes, with appropriate lane numbers and layouts.
- Correct lane-to-lane connections and turning movements at all junctions—not only at major intersections.
- All major, minor, secondary, local, and neighbourhood roads, as well as any highways visible within the assigned GIS map.

Do not limit your model to one intersection or only the major roads. All roads and junctions within the map must be represented, except for the exclusions below.

- Parking spaces and driveways do not need to be modelled.

The objective is to develop a complete road network whose geometry, lane layout, and connections match the provided map as closely as reasonably possible.

Stage A1 focuses on road-network development. Traffic signal implementation, traffic flow, right-of-way settings, speed limits, and scenario development will be addressed in Stage A2.

Submission files:

1. Create a folder named with your group number, for example, Group1.
2. Include the following files, using the standard SUMO file extensions shown below.

Required file	Example filename
SUMO network file	`Group1.net.xml`
NETEDIT configuration file	`Group1.netecfg`
SUMO simulation configuration file	`Group1.sumocfg`

3. Do not include the GIS image in your submission.
4. Compress your group folder into a single ZIP file, such as Group1.zip, and have one group member submit it through Avenue on behalf of the group.

Provided Files and Map Setup:

Before developing your road network, read your group's TXT file. It contains:

- Map width (W) and height (H): Enter these values in Edit Visualization when setting up your GIS background map.
- Real-world location coordinates: These are provided for geographic reference, to help you identify the assigned study area, connections, etc.

Use the GIS image and setup values provided specifically for your group. Set up the background map before beginning your road-network development.

Stage A2 - Urban Roadway Simulation in SUMO

Purpose and Assignment Sequence

Stage A2 continues directly from the Stage A1 network. Complete the Stage A1 digital roadway network first, then identify the assigned intersection (GroupX.pptx) and configure major-road traffic operations, develop the required simulation conditions, and evaluate your selected roadway edge. Stage A2 and Report A are submitted together and are worth 10% of the final course grade. Then, you will conduct:

Stage A2 Milestones

1. Traffic Flow: Configure traffic demand and routes on the major roads of the Stage A1 network **(only the selected intersection - the start of each traffic flow is clear based on the GroupX.pptx).**

2. Traffic Signal: Implement the applicable traffic-signal timing on the intersection (you may assume same values for each signal phasing as we did for lab 2). Otherwise, you need to write your assumptions.

3. Adjust Right of Way: Check and adjust junction priorities and right-of-way behaviour so that intended traffic movements operate reasonably.

4. Simulate HDVs and CAVs: Develop comparable 100% HDV and 100% CAV conditions. Keep the network, demand, routes, signal timing, observation period, and simulation duration consistent except for the intended vehicle-behaviour/composition change.

5. Intervention: Select one justified intervention, implement it using the 100% HDV baseline, and evaluate its effect on the same selected major-road edge. The intervention must directly affect the same street selected for your speed–density analysis.

Required Simulation Conditions:

1. Base Scenario: 100% human-driven vehicles (HDVs). This establishes the baseline traffic condition.

2. Scenario 1: 100% autonomous vehicles (CAVs). Only the intended HDV-to-CAV vehicle characteristics/composition should change.

3. Scenario 2: 100% HDVs plus ONE selected intervention (X). Examples include a lane-configuration change, major-road signal-timing adjustment, or a justified traffic-demand adjustment.

Controlled-comparison rule: Hold all unrelated inputs constant across the three conditions. The purpose is to isolate the intended CAV or intervention change rather than create three unrelated models.

Required Performance Analysis

- Select ONE representative upstream edge on a major road, **immediately approaching the intersection** (where vehicles travel toward the intersection). Use the SAME edge in all three scenarios for a consistent comparison.
- Speed–Density is the ONLY required fundamental diagram for Mini Project A.
- Produce three Speed–Density diagrams for the selected edge: Base Scenario, Scenario 1 (100% CAV), and Scenario 2 (Intervention).
- Do not assume that CAVs or the intervention will improve performance. Explain what the simulation actually shows and identify limitations.

Stage A2 Evidence to Keep for Report A

- A concise table of Stage A2 baseline assumptions, including traffic demand, routes/turning assumptions, signal timing, speed limits, simulation duration, and edgeData aggregation period where applicable.
- The selected edge identification and the three Speed–Density diagrams.
- Documentation of the intervention, including the before/after parameter values and at least one implementation screenshot.
- Notes explaining the main differences between the HDV, CAV, and intervention results and any limitations of the simulation

Report A - Integrated Stage A1 and A2 Municipal Case Study

After completing both project stages, submit ONE integrated industry report-style Report A of approximately 5,000-6,000 words. For this hypothetical classroom task, act as a traffic-engineering consulting team helping the City of Toronto understand the assigned Intersection X-Y, establish the simulated existing condition, explore a 100% CAV future, and investigate ONE candidate intervention.

Report A must integrate Stage A1 and Stage A2 rather than treating them as separate reports. The three study problems/objectives should remain aligned throughout the Introduction, Literature Review, Methodology, Result Analysis, and Conclusion: (1) develop the roadway-and-traffic baseline; (2) test the 100% CAV effect on the selected edge; and (3) test one intervention on that same edge.

Required Preparation for Report A - Generative AI Writing Videos

Before beginning the Report A writing process, every group should watch both videos below. The videos demonstrate a workflow for using generative AI to support research-paper/technical-report writing.

Video 1: https://youtu.be/DsGvDcmw5ds

Video 2: https://youtu.be/21tJPYt951s

After watching the vidoes, download **the below provided word/ppt file in Avenue** and start writing:

1. WritingStructure.docx
2. Template.docx
3. GroupX.pptx (for Figures samples)

Responsible AI requirement: Generative AI is a writing/research-support tool, not a substitute for the group's simulation work or engineering judgment. Do not invent data, simulation results, City of Toronto claims, or references. Verify all technical statements, numerical values, interpretations, and cited sources before submission, and include an honest AI-use statement in the report.

Do not submit AI-generated content "as is." Use it only as a starting point, then review, verify, revise, and rewrite it in your own words as appropriate.

Submission Files - Stage A2 + Report A

Create a folder named with your group number, for example, **Group1**. Your submission must contain the complete SUMO files required to reproduce the three scenarios developed in Stage A2, together with the final Report A.

Required item	**Example**
Base Scenario - 100% HDV	Base_Scenario/
Scenario 1 - 100% CAV	Scenario_1_CAV/
Scenario 2 - Intervention	Scenario_2_Intervention/
Report A - PDF version	Group1_ReportA.pdf
Report A - Word version	Group1_ReportA.docx

Each SUMO scenario folder should contain all files required to open and run that scenario, including, as applicable:

- SUMO network file (.net.xml)
- NETEDIT configuration file (.netecfg)
- SUMO simulation configuration file (.sumocfg)
- Route/traffic-demand files (.rou.xml)
- Additional files (.add.xml)

- Generated **edgeData / Speed–Density XML output** used for the analysis (DiagramsScenario.xml)

The three scenarios must correspond to:

Base Scenario: 100% Human-Driven Vehicles (HDVs)
Scenario 1: 100% Autonomous Vehicles (CAVs)
Scenario 2: 100% HDVs + ONE selected intervention

Use the **same selected major-road edge** for the Speed–Density analysis in all three scenarios.

Report A
Submit Report A in **both formats**:

- Group1_ReportA.pdf - final version used for grading
- Group1_ReportA.docx - editable Word version

The PDF and Word files must contain the **same final report content**.

Final Submission

- Place all scenario folders and both Report A files inside the Group1 folder. Then compress the complete folder into one ZIP file:
- **Group1.zip**
- Only **one group member** should submit the ZIP file through Avenue on behalf of the group.
- **Do not include the original GIS image** in the submission.